

2025 AMC 10A Problem 14 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10A Problem 14. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10A Problem 14

Problem:

Six chairs are arranged around a round table. Two students and two teachers randomly select four of the chairs to sit in. What is the probability that the two students will sit in two adjacent chairs and the two teachers will also sit in two adjacent chairs?

Answer Choices:

A. $\frac{1}{6}$

B. $\frac{1}{5}$

C. $\frac{2}{9}$

D. $\frac{3}{13}$

E. $\frac{1}{4}$

Solution:

If we look around the circle, we can seat the people in $6 \cdot 5 \cdot 4 \cdot 3$ total ways (with full distinguishability).

However, if our condition is met, we will first seat the teachers in $6 \cdot 2$ ways (the second teacher must occupy one of the two adjacent spots).

Then, the remaining students can choose their seats in 6 ways (there are 4 chairs left, so there are 3 pairs of adjacent seats, and 2 orders for them to sit in).

Thus our overall probability is

$$\frac{6 \cdot 2 \cdot 6}{6 \cdot 5 \cdot 4 \cdot 3} = \frac{1}{5}$$

which gives us an answer of

$$\boxed{(\mathbf{B}) \frac{1}{5}}$$

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